

Leonor N. L. Simões

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Summary

I am a highly motivated 2nd year PhD candidate in Astrophysics at University College London. I have a strong interest in Cosmology, particularly in the study of the basic properties and evolution of the large-scale structure in the Universe and how we can combine observational and theoretical techniques to test and refine the cosmological model.

Experience

Euclid SGS Support Scientist

University College London • London, United Kingdom.

06/2024 - 09/2024

Over the summer, I joined the Euclid team in the UCL Astrophysics Group.

MSci Project

University College London • London, United Kingdom

09/2023 - 06/2024

My MSci project explored the properties of the cosmic web. Using Python, I investigated how the cosmic web differs between cosmological models, using N-body simulations, Hessian-based Cosmic Web classification methods, and the Minimum Spanning Tree. Supervised by Dr Krishna Naidoo and Prof. Benjamin Joachimi. Culminated in a written report and a presentation, in which I obtained first-class honours for both.

Brian Duff Astro Undergraduate Summer Research Studentship

University College London • London, United Kingdom

06/2023-09/2023

A summer internship supervised by Dr Krishna Naidoo and Prof. Ofer Lahav on a computational project exploring the relationship between the Minimum Spanning Tree and the Cosmic Web. Culminated with an in-person presentation.

AstroCamp 2019

Centre for Astrophysics of the University of Porto • Porto, Portugal

08/2019-08/2019

This was a highly selective two-week astrophysics summer programme, in English, for students interested in studying astrophysics at university. We worked on a computational research project, using Python to analyse data from supernovae to motivate the existence of dark energy.

Education

PhD Astrophysics

University College London • London, United Kingdom

2024 - Now

Supervised by Prof. Benjamin Joachimi, I am working on intrinsic alignments using data from the Euclid telescope and the DESI survey.

MSci Astrophysics

University College London • London, United Kingdom

Finished 2024

Completed with first-class honours.

Awards

- Herschel prize (2024) - Best performance by a 4th year Astrophysics student.
- Clockmakers' prize (2023) - UCL Observatory student of the year award.
- Brian Duff Astro Undergraduate Summer Research Studentship (2023) - Bursary available to undergraduates interested in pursuing research projects at UCL over the summer.
- Huggins Prize (2022) - Best performance 2nd year astrophysics.

Publications

- 'How massive neutrinos reshape the cosmic web.' [LNLS et al. 2024](#)

Teaching assistant

- Practical Astronomy 1 – 2025/26
3rd year module covering observational astrophysics at the UCL Observatory.
- Physical Cosmology – 2025/26
3rd module introducing basic cosmological principles and a summary of selected topics in extragalactic astronomy.